

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 04

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Specialization	CSNE
Batch No	Y2S2.WD.CSNE.01

Activity 01

```
GNU nano 7.2 act1.c
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("My process ID is: %d\n", getpid());
    return 0;
}
```

a)

```
vboxuser@ubuntu: ~
vboxuser@ubuntu:~$ nano act1.c
vboxuser@ubuntu:~$ gcc -oact1 act1.c
vboxuser@ubuntu:~$ ./act1
My process ID is: 3505
```

b) getpid()

c) getpid() returns the process ID (PID) of the currently running program.

d) When <unistd.h> is removed, the compiler cannot recognize the When getpid() function because it is declared in that header file. Therefore, the program gives a compilation error or warning.

Activity 02

```
GNU nano 7.2 act1.c
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("My process ID is: %d\n", getpid());
    printf("My parent process ID is: %d\n", getppid());
    return 0;
}
```

```
vboxuser@ubuntu:~$ nano act1.c
vboxuser@ubuntu:~$ gcc -oact1 act1.c
vboxuser@ubuntu:~$ ./act1
My process ID is: 4447
My parent process ID is: 3051
```

Activity 03

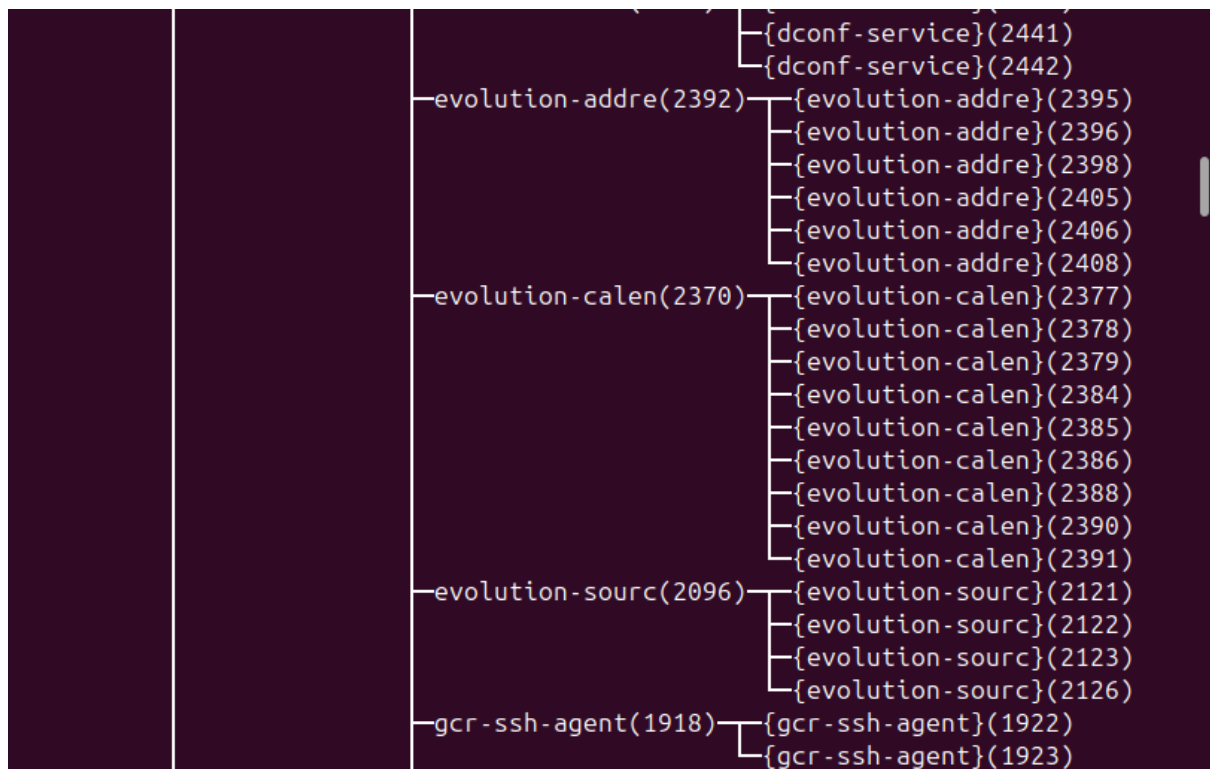
```
vboxuser@ubuntu: ~
GNU nano 7.2 act1.c
#include <stdio.h>
#include <unistd.h>

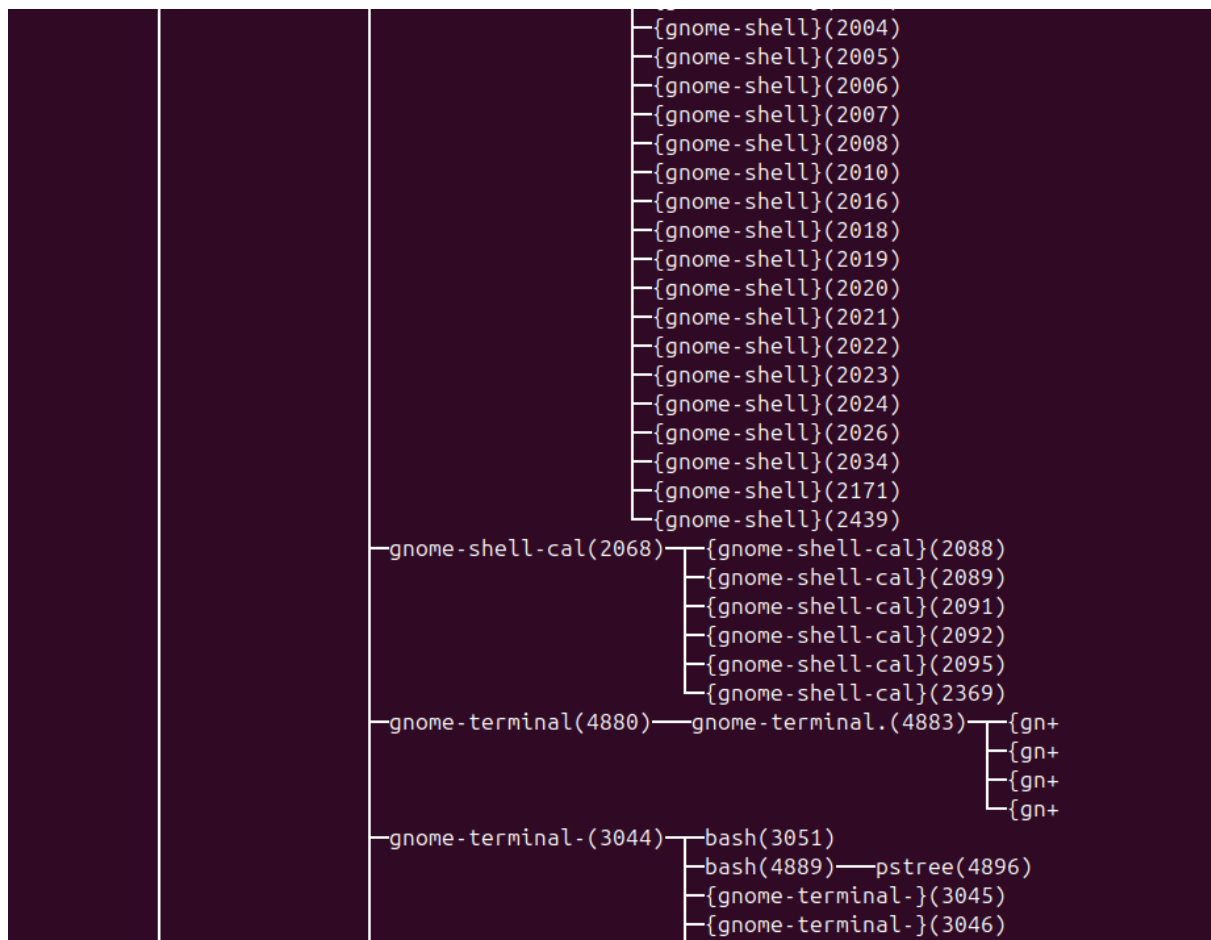
int main() {
    printf("My process ID is: %d\n", getpid());
    printf("My parent process ID is: %d\n", getppid());
    sleep(30);
    return 0;
}
```

```
vboxuser@ubuntu:~$ nano act1.c
vboxuser@ubuntu:~$ gcc -oact1 act1.c
vboxuser@ubuntu:~$ ./act1
My process ID is: 4481
My parent process ID is: 3051
```

```
vboxuser@ubuntu: ~  
vboxuser@ubuntu:~$ pstree -p  
systemd(1)─ModemManager(940)─{ModemManager}(980)  
├─{ModemManager}(996)  
└─{ModemManager}(999)  
├─NetworkManager(877)─{NetworkManager}(1001)  
├─{NetworkManager}(1004)  
└─{NetworkManager}(1005)  
├─accounts-daemon(798)─{accounts-daemon}(836)  
├─{accounts-daemon}(837)  
└─{accounts-daemon}(839)  
├─avahi-daemon(737)─avahi-daemon(871)  
├─colord(1369)─{colord}(1374)  
├─{colord}(1375)  
└─{colord}(1377)  
├─cron(800)  
├─cups-browsed(1130)─{cups-browsed}(1169)  
├─{cups-browsed}(1170)  
└─{cups-browsed}(1171)  
├─cupsd(1086)  
├─dbus-daemon(739)  
└─fwupd(4015)─{fwupd}(4016)  
├─{fwupd}(4017)  
├─{fwupd}(4018)  
└─{fwupd}(4019)
```

```
└─{fwupd}(4021)  
├─gdm3(1110)─gdm-session-wor(1728)─gdm-wayland-ses(1823)─gnome+  
├─{gdm-session-wor}(1729)  
├─{gdm-session-wor}(1730)  
└─{gdm-session-wor}(1731)  
├─{gdm3}(1122)  
├─{gdm3}(1123)  
└─{gdm3}(1125)  
├─gnome-remote-de(750)─{gnome-remote-de}(985)  
├─{gnome-remote-de}(988)  
└─{gnome-remote-de}(990)  
├─kerneloops(1155)  
├─kerneloops(1159)  
├─polkitd(772)─{polkitd}(896)  
├─{polkitd}(898)  
└─{polkitd}(900)  
├─power-profiles-(784)─{power-profiles-}(808)  
├─{power-profiles-}(809)  
└─{power-profiles-}(825)  
└─rsyslogd(870)─{rsyslogd}(904)
```





```

      {gnome-terminal-}(3048)
      {gnome-terminal-}(3049)
      {gnome-terminal-}(3050)
goa-daemon(2347)  {goa-daemon}(2350)
                  {goa-daemon}(2351)
                  {goa-daemon}(2353)
                  {goa-daemon}(2354)
goa-identity-se(2357)  {goa-identity-se}(2361)
                      {goa-identity-se}(2362)
                      {goa-identity-se}(2364)
gsd-a11y-settin(2101)  {gsd-a11y-settin}(2136)
                      {gsd-a11y-settin}(2137)
                      {gsd-a11y-settin}(2158)
                      {gsd-a11y-settin}(2221)
gsd-color(2102)  {gsd-color}(2200)
                 {gsd-color}(2201)
                 {gsd-color}(2204)
                 {gsd-color}(2254)
gsd-datetime(2103)  {gsd-datetime}(2170)
                   {gsd-datetime}(2172)
                   {gsd-datetime}(2197)
                   {gsd-datetime}(2298)
gsd-housekeepin(2104)  {gsd-housekeepin}(2150)
                      {gsd-housekeepin}(2151)
                      {gsd-housekeepin}(2164)
                      {gsd-housekeepin}(2188)
gsd-keyboard(2105)  {gsd-keyboard}(2139)
                   {gsd-keyboard}(2140)
                   {gsd-keyboard}(2148)
                   {gsd-keyboard}(2161)
gsd-media-keys(2106)  {gsd-media-keys}(2176)
                     {gsd-media-keys}(2177)
                     {gsd-media-keys}(2182)

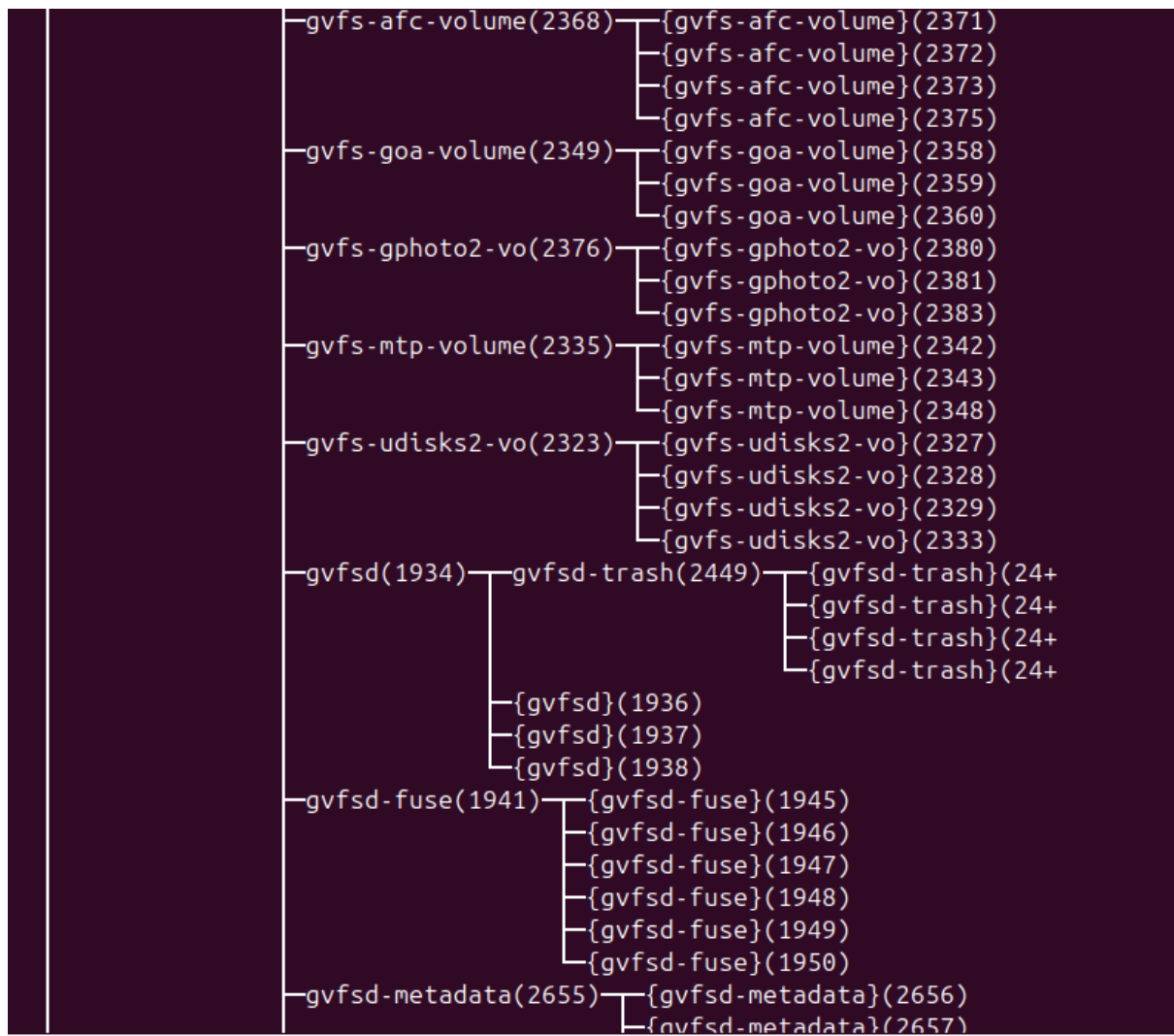
```



```

└─{gsd-media-keys}(2215)
gsd-power(2107)└─{gsd-power}(2146)
               └─{gsd-power}(2149)
               └─{gsd-power}(2156)
               └─{gsd-power}(2163)
gsd-print-notif(2112)└─{gsd-print-notif}(2249)
                    └─{gsd-print-notif}(2250)
                    └─{gsd-print-notif}(2253)
gsd-printer(2313)└─{gsd-printer}(2318)
                 └─{gsd-printer}(2319)
                 └─{gsd-printer}(2320)
gsd-rfkill(2113)└─{gsd-rfkill}(2129)
                └─{gsd-rfkill}(2130)
                └─{gsd-rfkill}(2154)
gsd-screensaver(2114)└─{gsd-screensaver}(2174)
                    └─{gsd-screensaver}(2175)
                    └─{gsd-screensaver}(2247)
gsd-sharing(2116)└─{gsd-sharing}(2133)
                 └─{gsd-sharing}(2134)
                 └─{gsd-sharing}(2135)
                 └─{gsd-sharing}(2159)
gsd-smartcard(2117)└─{gsd-smartcard}(2162)
                   └─{gsd-smartcard}(2169)
                   └─{gsd-smartcard}(2192)
                   └─{gsd-smartcard}(2297)
gsd-sound(2124)└─{gsd-sound}(2142)
                └─{gsd-sound}(2143)
                └─{gsd-sound}(2160)
                └─{gsd-sound}(2189)
gsd-wacom(2125)└─{gsd-wacom}(2179)
                └─{gsd-wacom}(2180)
                └─{gsd-wacom}(2190)
                └─{gsd-wacom}(2241)

```

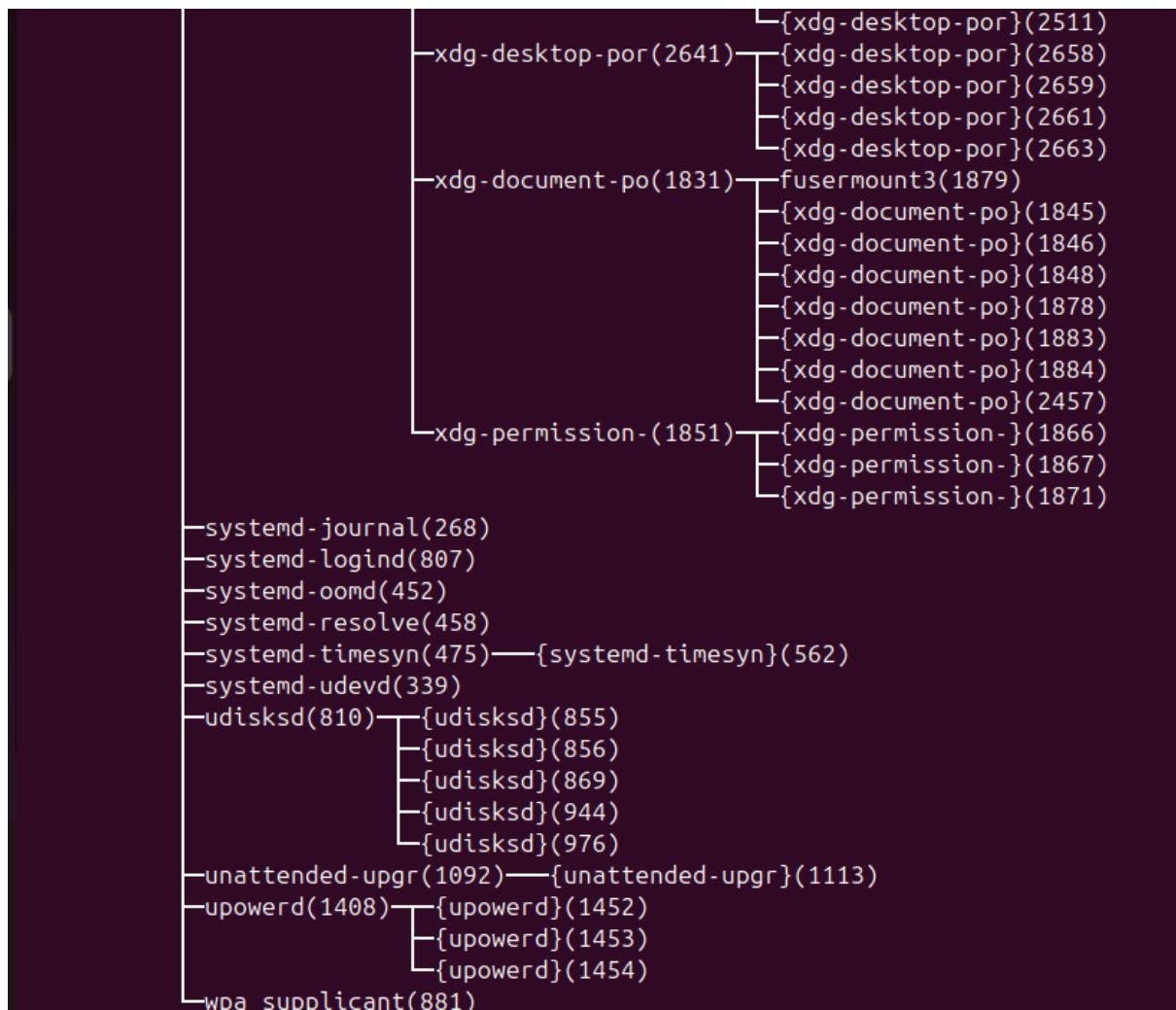


		└─{gvfsd-metadata}(2662)
	ibus-daemon(2100)	ibus-dconf(2277)
		└─{ibus-dconf+}
		└─{ibus-dconf+}
		└─{ibus-dconf+}
		└─{ibus-dconf+}
		ibus-engine-sim(2400)
		└─{ibus-+}
		└─{ibus-+}
		└─{ibus-+}
		ibus-extension-(2280)
		└─{ibus-+}
		└─{ibus-+}
		└─{ibus-+}
		└─{ibus-+}
		{ibus-daemon}(2233)
		{ibus-daemon}(2234)
		{ibus-daemon}(2251)
	ibus-portal(2285)	{ibus-portal}(2288)
		{ibus-portal}(2289)
		{ibus-portal}(2293)
	pipewire(1754)	{pipewire}(1788)
		{pipewire}(1800)
	pipewire(1755)	{pipewire}(1796)
		{pipewire}(1797)
	pipewire-pulse(1760)	{pipewire-pulse}(1787)
		{pipewire-pulse}(1794)
	snapd-desktop-i(2509)	snapd-desktop-i(2639)
		└─{s+}
		└─{s+}
		└─{s+}
		└─{s+}
	tracker-miner-f(2481)	{tracker-miner-f}(2484)
		{tracker-miner-f}(2485)
		{tracker-miner-f}(2487)
		{tracker-miner-f}(2493)

```

-pipewire(1754)---{pipewire}(1788)
                  {pipewire}(1800)
-pipewire(1755)---{pipewire}(1796)
                  {pipewire}(1797)
-pipewire-pulse(1760)---{pipewire-pulse}(1787)
                       {pipewire-pulse}(1794)
-snapd-desktop-i(2509)---snapd-desktop-i(2639)---{s+
                                                  {s+
                                                  {s+
                                                  {s+
-tracker-miner-f(2481)---{tracker-miner-f}(2484)
                        {tracker-miner-f}(2485)
                        {tracker-miner-f}(2487)
                        {tracker-miner-f}(2493)
                        {tracker-miner-f}(2499)
                        {tracker-miner-f}(2506)
                        {tracker-miner-f}(2507)
-wireplumber(1758)---{wireplumber}(1789)
                   {wireplumber}(1790)
                   {wireplumber}(1792)
                   {wireplumber}(1802)
                   {wireplumber}(1807)
-xdg-desktop-por(2483)---{xdg-desktop-por}(2490)
                       {xdg-desktop-por}(2492)
                       {xdg-desktop-por}(2494)
                       {xdg-desktop-por}(2638)
                       {xdg-desktop-por}(2677)
                       {xdg-desktop-por}(2678)
-xdg-desktop-por(2498)---{xdg-desktop-por}(2503)
                       {xdg-desktop-por}(2504)
                       {xdg-desktop-por}(2505)
                       {xdg-desktop-por}(2510)
                       {xdg-desktop-por}(2511)

```



Activity 04

```
GNU nano 7.2 act1.c
#include <stdio.h>
#include <stdlib.h>

int main() {
    printf("Before exit\n");
    exit(0);
    printf("This line will not be executed\n");
    return 0;
}
```

```
vboxuser@ubuntu:~$ nano act1.c
vboxuser@ubuntu:~$ gcc -oact1 act1.c
vboxuser@ubuntu:~$ ./act1
Before exit
```

Explanation:

exit(0) terminates the program immediately.

So the next printf() line never executes.

Activity 05

```
GNU nano 7.2 act1.c
#include <stdio.h>
#include <unistd.h>

int main() {
    fork();
    printf("Hello world\n");
    return 0;
}
```

```
vboxuser@ubuntu:~$ nano act1.c
vboxuser@ubuntu:~$ gcc -oact1 act1.c
vboxuser@ubuntu:~$ ./act1
Hello world
Hello world
```

Explanation:

fork() creates a child process.

After fork(), both:

- Parent process
- Child process

execute the next line:

```
printf("Hello world\n");
```

So it prints twice.

Activity 06

```
GNU nano 7.2 act1.c
#include <stdio.h>
#include <unistd.h>

int main() {
    int ret = fork();

    printf("Return value of fork(): %d\n", ret);

    if (ret == 0) {
        printf("Child process ID: %d\n", getpid());
        printf("Child's parent process ID: %d\n", getppid());
    }
    else {
        printf("Parent process ID: %d\n", getpid());
    }

    sleep(20);

    return 0;
}
```

```
vboxuser@ubuntu:~$ nano act1.c
vboxuser@ubuntu:~$ gcc -oact1 act1.c
vboxuser@ubuntu:~$ ./act1
Return value of fork(): 4618
Parent process ID: 4617
Return value of fork(): 0
Child process ID: 4618
Child's parent process ID: 4617
```

Activity 07

```
GNU nano 7.2 act1.c
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Before execl()\n");

    execl("/bin/ls", "ls", "-l", NULL);

    printf("This will not print if execl works\n");

    return 0;
}
```

```

vboxuser@ubuntu:~$ nano act1.c
vboxuser@ubuntu:~$ gcc -oact1 act1.c
vboxuser@ubuntu:~$ ./act1
Before execl()
total 84
-rwxrwxr-x 1 vboxuser vboxuser 16000 Feb 18 17:01 act1
-rw-rw-r-- 1 vboxuser vboxuser 188 Feb 18 17:01 act1.c
-rwxrwxr-x 1 vboxuser vboxuser 16208 Feb 15 05:11 activity3
-rw-rw-r-- 1 vboxuser vboxuser 791 Feb 15 05:11 activity3.c
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Downloads
-rw-rw-r-- 1 vboxuser vboxuser 230 Feb 8 08:42 first.c
-rw-rw-r-- 1 vboxuser vboxuser 1 Feb 8 08:39 first.c.save
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Public
drwx----- 6 vboxuser vboxuser 4096 Feb 8 08:21 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Videos

```

Explanation

- execl() replaces the current process program with ls
- If execl works, it never comes back
- So the last printf will not execute

Activity 08

(A) Original system() Program

```

GNU nano 7.2 act1.c
#include <stdio.h>
#include <stdlib.h>

int main() {
    printf("Before system call\n");
    system("ls -l");
    printf("After system call\n");

    return 0;
}

```



```

vboxuser@ubuntu:~$ nano act1.c
vboxuser@ubuntu:~$ nano act1.c
vboxuser@ubuntu:~$ gcc -oact1 act1.c
vboxuser@ubuntu:~$ ./act1
Before system call
total 84
-rwxrwxr-x 1 vboxuser vboxuser 16000 Feb 18 17:53 act1
-rw-rw-r-- 1 vboxuser vboxuser 154 Feb 18 17:53 act1.c
-rwxrwxr-x 1 vboxuser vboxuser 16208 Feb 15 05:11 activity3
-rw-rw-r-- 1 vboxuser vboxuser 791 Feb 15 05:11 activity3.c
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Downloads
-rw-rw-r-- 1 vboxuser vboxuser 230 Feb 8 08:42 first.c
-rw-rw-r-- 1 vboxuser vboxuser 1 Feb 8 08:39 first.c.save
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Public
drwx----- 6 vboxuser vboxuser 4096 Feb 8 08:21 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Videos
After system call

```

Explanation:

system("ls -l") runs the command in a shell and then returns back.

(B) Modified Program with fork()

```

GNU nano 7.2 act1.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
int main() {
    fork();
    printf("Before system()\n");
    system("ls -l");
    printf("After system()\n");

    return 0;
}

```

```

vboxuser@ubuntu:~$ nano act1.c
vboxuser@ubuntu:~$ gcc -oact1 act1.c
vboxuser@ubuntu:~$ ./act1
Before system()
Before system()
total 84
-rwxrwxr-x 1 vboxuser vboxuser 16040 Feb 18 18:00 act1
-rw-rw-r-- 1 vboxuser vboxuser 177 Feb 18 18:00 act1.c
-rwxrwxr-x 1 vboxuser vboxuser 16208 Feb 15 05:11 activity3
-rw-rw-r-- 1 vboxuser vboxuser 791 Feb 15 05:11 activity3.c
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Downloads
-rw-rw-r-- 1 vboxuser vboxuser 230 Feb 8 08:42 first.c
-rw-rw-r-- 1 vboxuser vboxuser 1 Feb 8 08:39 first.c.save
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Public
drwx----- 6 vboxuser vboxuser 4096 Feb 8 08:21 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Videos
total 84
-rwxrwxr-x 1 vboxuser vboxuser 16040 Feb 18 18:00 act1
-rw-rw-r-- 1 vboxuser vboxuser 177 Feb 18 18:00 act1.c
-rwxrwxr-x 1 vboxuser vboxuser 16208 Feb 15 05:11 activity3
-rw-rw-r-- 1 vboxuser vboxuser 791 Feb 15 05:11 activity3.c
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Downloads
-rw-rw-r-- 1 vboxuser vboxuser 230 Feb 8 08:42 first.c
-rw-rw-r-- 1 vboxuser vboxuser 1 Feb 8 08:39 first.c.save
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Public
drwx----- 6 vboxuser vboxuser 4096 Feb 8 08:21 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Jan 30 10:52 Videos
After system()
After system()

```

Explanation:

- fork() creates child process
- Both parent and child run system("ls -l")
- So the command executes twice